Linux Commands

1. lsblk – Command explanation

Command: lsblk

Purpose: Lists block devices such as disks, partitions, and swap devices in a tree-like format.

Observed result: The screenshot listed sda (388.4M), sdb (186M), sdc (2G, marked [SWAP]), and sdd (1T, mounted for the WSL distribution). Device names and sizes vary by system.

2. findmnt – Command explanation

Command: findmnt

Purpose: Displays mounted filesystems, showing mount target, source, filesystem type, and mount options.

Observed result: The screenshot showed / mounted from /dev/sdd using ext4, plus WSL-related mounts such as /mnt/wsl, /dev, and /sys. These special mounts are normal in WSL2.

3. du – Command explanation

Command: du -sh ~/srini1

Purpose: Shows the total disk space used by the srini1 directory. -s gives one summary; -h uses readable units.

Observed result: The output was 4.0K for /home/Srini/srini1.

4. df – Command explanation

Command: df -i

Purpose: Displays inode capacity and usage for mounted filesystems. Inodes store file metadata; a filesystem can run out of inodes even if disk space remains.

Observed result: The screenshot showed inode usage for the WSL ext4 filesystem and special mounts. Many entries showed about 1% use. Some Windows/DrvFS-style mounts may show unavailable or unusual inode figures.

5. grep – Command explanation

Command: grep -n "Hi" file1.txt

Purpose: Searches file1.txt for Hi. The -n option displays the matching line number.

Observed result: Output: 1:Hi, meaning Hi was found on line 1.

6. grep – Command explanation

Command: grep -r "are" ~/ --include="*.txt"

Purpose: Searches recursively under the home directory for are, limiting the search to files ending in .txt.

Observed result: Matches appeared in file1.txt, names.txt, and file2.txt. Results depend on the current file contents.

7. cut – Command explanation

Command: cut -d' ' -f1 file1.txt

Purpose: Uses a space as the delimiter and prints the first field from each line. It works best when fields are consistently separated.

Observed result: The screenshot displayed the first field from each line, such as Hi, Hello, How, are, and you?. Multiple spaces can affect field selection.

8. sort – Command explanation

Command: sort file1.txt

Purpose: Prints the lines of file1.txt in sorted order. It does not modify the original file.

Observed result: The lines appeared alphabetically as Hello, Hi, How, are, followed by the line beginning you?.

9. umask – Command explanation

Command: umask -S

Purpose: Displays the current default permission mask in symbolic form. The mask controls which permissions are removed when new files and directories are created.

Observed result: Output: u=rwx,g=rx,o=rx. This is the current mask representation, not the permissions of an existing file.

10. sudo – Command explanation

Command: sudo chgrp root file1.txt

Purpose: Changes the group owner of file1.txt to root. sudo runs the command with elevated privileges.

Observed result: The following ls -l output showed owner Srini and group root, confirming the group change.

11. ls – Command explanation

Command: ls -l file1.txt

Purpose: Displays a long listing with permissions, link count, owner, group, file size, modification time, and filename.

Observed result: Output showed -rw-rw-r--, owner Srini, group root, size 46 bytes, and file1.txt.

12. grep – Command explanation

Command: grep -n "Linux" file1.txt

Purpose: Searches file1.txt for Linux and would display line numbers for matches.

Observed result: No output appeared, which means the exact text Linux was not found in the file. grep normally returns status 1 when there is no match; this is not necessarily an error.